

# ADIDAS US SALES SQL ANALYSIS

Darryl Yuma Ardhika | [LinkedIn](#) | [GitHub](#) | [Dataset \(Kaggle\)](#)

## Project Overview

This project analyzes Adidas sales performance across the United States using SQL (SQLite). The mock dataset used for this project contains information on retailers, sales regions, products, units sold, total sales, operating margins, and sales methods.

The objective of this analysis is to identify factors that drive sales revenue and profitability. By analyzing the available metrics, this project aims to uncover patterns and trends that may be used to support future business decisions.

Key areas of focus for this analysis include product profitability, regional performance, sales channel effectiveness, retailer-product performance, and sales trends over time.

## Findings

- Men's Street Footwear is the product that contributes the most to total revenue at 23% with Women's Athletic Footwear contributing the least at 11%. Contribution percentage across other products are relatively balanced ranging from 13% to 19%.
- Revenue split between men's and women's products is relatively balanced with a 53/47 revenue split. This suggests that Adidas maintains an equal customer base across both segments.
- Apparel products consistently outperform footwear products in profit per unit sold. Women's Apparel and Men's Apparel exceed the average by 17.53% and 9.53% respectively, while all three below average categories are footwear products, suggesting that footwear sales are less profitable on a per-unit basis.
- The West and Northeast regions generate low profit margins (35.73% and 38.81%) despite generating high revenue (\$36.4M and \$25M). Inversely, the South and Midwest regions generate high profit margins (44.76% and 41.14%) despite generating low revenue (\$20.6M and \$16.6M).
- Sports focused retailers significantly outperform general retailers in both profit and revenue. Retailers like Foot Locker, Sports Direct, and West Gear generate over double the amount of profit and revenue compared to retailers like Kohl's, Amazon, and Walmart.
- The company experienced significant growth from 2020 to 2021 with Year over Year revenue growth increasing by 295.79% and profit growth increasing by 323.82%.
- Online sales are the strongest sales channel contributing to 37.42% of total revenue and having the highest profit margin at 43.48%.
- July, August, and December (the summer months and the year end holidays) are when revenue and profits are at their peak. Meanwhile February, March, October, and November are when revenue and profits are at their lowest.

## Recommendations

- Revisiting marketing strategies for underperforming products such as Women's Athletic Footwear should be considered to close the gap with other product categories. Strategies such as targeted campaigns or promotional pricing are one such example.
- Since sports focused retailers generate over double the revenue and profit of general retailers, inventory allocation and marketing prioritization should be considered while assessing the value of maintaining a broad presence in general retail.
- Operating cost structures in the West and Northeast regions should be investigated such as distribution and retailer costs and renegotiating terms to bring profit margins closer to company average to improve overall profitability.
- Continue investing into online sales channels by improving e-commerce experience and digital marketing as it is currently the highest revenue and profit driver at 43.48% and 37.42% respectively.
- Cross selling strategies should be explored such as bundling apparel with footwear ("complete your look" type of promotion) to shift a greater share of sales towards higher profit margin products.
- Marketing efforts and inventory planning should be intensified during peak demand months like July, August, and December to align with seasonal increases in demand.
- Promotion campaigns and targeted discounts should be considered during weaker months such as February-March and October-November to stimulate demand and improve sales performance.

## Questions and SQL Query List

| No | Question                                            | Query                                                                                                                                                                                                                                                               | Answer                                                                                                                       |
|----|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1  | Which product contributes the most to total sales?  | <pre>SELECT     Product,     ROUND(SUM(Total_Sales), 2) AS     Product_Total,     ROUND((100*SUM(Total_Sales)) / (SELECT     SUM(Total_Sales) FROM Sales), 2) AS     Contribution_Percent FROM     Sales GROUP BY     Product ORDER BY     Product_Total DESC</pre> | <ul style="list-style-type: none"><li>• Men's street footwear contributes most to total sales with a share of 23%.</li></ul> |
| 2  | What product has the highest profit per units sold? | <pre>WITH metrics AS (SELECT     Product,     SUM(Units_Sold) AS total_units_sold,</pre>                                                                                                                                                                            | <ul style="list-style-type: none"><li>• Women's Apparel has the highest profit per unit sold with</li></ul>                  |

|   |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                              |
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|   |                                                                                                    | SUM(Operating_Profit) AS<br>product_total_profit<br>FROM<br>Sales<br>GROUP BY<br>Product)<br>SELECT<br>Product,<br>ROUND(product_total_profit / total_units_sold, 2)<br>AS profit_per_unit<br>FROM<br>metrics<br>ORDER BY<br>profit_per_unit DESC                                                                                                                                                                                                                                                                               | 22.33 profit per unit.                                                                                                                                                                                                                                       |
| 3 | Which region contributes the most to total sales?                                                  | SELECT<br>Region,<br>ROUND(SUM(Total_Sales), 2) AS<br>Region_Total,<br>ROUND((100.0*SUM(Total_Sales))/(SELECT<br>SUM(Total_Sales) FROM Sales), 2) AS<br>Contribution_Percent<br>FROM<br>Sales<br>GROUP BY<br>Region<br>ORDER BY<br>Region_Total DESC                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>The West region contributes the most to total sales with 30.32% total.</li> </ul>                                                                                                                                     |
| 4 | How did Adidas' sales and operating profits change from 2020 to 2021, and what was the YoY growth? | WITH yearly_metrics AS(<br>SELECT<br>SUM(CASE WHEN strftime('%Y',<br>Invoice_Date) = '2020' THEN Total_Sales ELSE 0<br>END) AS sales_2020,<br>SUM(CASE WHEN strftime('%Y',<br>Invoice_Date) = '2021' THEN Total_Sales ELSE 0<br>END) AS sales_2021,<br>SUM(CASE WHEN strftime('%Y',<br>Invoice_Date) = '2020' THEN Operating_Profit ELSE<br>0 END) AS profit_2020,<br>SUM(CASE WHEN strftime('%Y',<br>Invoice_Date) = '2021' THEN Operating_Profit ELSE<br>0 END) AS profit_2021<br>FROM<br>Sales<br>)<br>SELECT<br>sales_2020, | <ul style="list-style-type: none"> <li>Sales from 2020 to 2021 increased from 24,237,235 to 95,929,325 with a YoY growth of 295.79%.</li> <li>Profits from 2020 to 2021 increased from 9,015,505.08 to 38,209,462.62 with a YoY growth of 323.82%</li> </ul> |

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|   |                                                                                                | sales_2021,<br>profit_2020,<br>profit_2021,<br>ROUND((100.0 * (sales_2021 - sales_2020) /<br>sales_2020),2) AS Sales_YoY_Growth,<br>ROUND((100 * (profit_2021 - profit_2020) /<br>profit_2020),2) AS Profit_YoY_Growth<br>FROM<br>yearly_metrics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                        |
| 5 | Which products fall below the average profit per unit, and how far below the average are they? | WITH metrics AS<br>(SELECT<br>Product,<br>SUM(Units_Sold) AS total_units_sold,<br>SUM(Operating_Profit) AS<br>product_total_profit,<br>ROUND(SUM(Operating_Profit) /<br>SUM(Units_Sold), 2) AS profit_per_unit<br>FROM<br>Sales<br>GROUP BY<br>Product),<br>avg_profit AS<br>(SELECT<br>ROUND(AVG(profit_per_unit),2) AS<br>avg_profit_per_unit<br>FROM<br>metrics)<br><br>SELECT<br>m.Product,<br>m.profit_per_unit,<br>a.avg_profit_per_unit,<br>ROUND(((m.profit_per_unit -<br>a.avg_profit_per_unit) / a.avg_profit_per_unit) * 100<br>,2) AS "Difference (Percentage)",<br>CASE WHEN a.avg_profit_per_unit ><br>m.profit_per_unit THEN 'Below Average' ELSE<br>'Above Average' END AS Performance<br>FROM<br>metrics AS m<br>CROSS JOIN<br>avg_profit AS a<br>ORDER BY<br>m.profit_per_unit DESC | <ul style="list-style-type: none"> <li>There are 3 products that fall below the average profit per unit (19.0) which are Women's Athletic Footwear (17.65), Men's Athletic Footwear (17.08), and Women's Street Footwear (16.56) being the worst performer.</li> </ul> |
| 6 | How do total revenue and                                                                       | SELECT<br>Retailer,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>The retailer-product</li> </ul>                                                                                                                                                                                                 |

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|   | operating profits look across retailer-product combinations ?       | Product,<br>SUM(Total_Sales) AS "Total Sales",<br>SUM(Operating_Profit) AS "Operating Profit"<br>FROM<br>Sales<br>GROUP BY<br>Retailer, Product<br>ORDER BY<br>"Operating Profit" DESC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | combination that generates the highest operating profit and also most revenue is Men's Street Footwear from Foot Locker                                                                                                                          |
| 7 | Which regions generate high sales but below-average profit margins? | WITH metrics AS<br>(SELECT<br>Region,<br>SUM(Total_Sales) AS total_region_sales,<br>ROUND(100 * SUM(Operating_Profit) /<br>SUM(Total_Sales), 2) AS region_profit_margin<br>FROM<br>Sales<br>GROUP BY<br>Region),<br>region_averages AS<br>(SELECT<br>ROUND(AVG(total_region_sales),2) AS<br>avg_region_sales,<br>ROUND(AVG(region_profit_margin),2) AS<br>avg_region_margin<br>FROM<br>metrics)<br><br>SELECT<br>m.Region,<br>m.total_region_sales,<br>m.region_profit_margin,<br>r.avg_region_sales,<br>r.avg_region_margin,<br>CASE WHEN m.total_region_sales ><br>r.avg_region_sales THEN 'Above Average' ELSE<br>'Below Average' END AS "Sales Performance",<br>CASE WHEN m.region_profit_margin ><br>r.avg_region_margin THEN 'Above Average' ELSE<br>'Below Average' END AS "Profit Margin<br>Performance",<br>CASE WHEN m.total_region_sales ><br>r.avg_region_sales AND m.region_profit_margin ><br>r.avg_region_margin THEN 'Strong Performer' | <ul style="list-style-type: none"> <li>• The West and Northeast regions generate high sales but below-average profit margins.</li> <li>• The South and Midwest regions inversely generate low sales but above-average profit margins.</li> </ul> |

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|   |                                                                                                       | <p>WHEN m.total_region_sales &gt;<br/>r.avg_region_sales AND m.region_profit_margin &lt;<br/>r.avg_region_margin THEN 'High Sales, Low Profitability'</p> <p>WHEN m.total_region_sales &lt;<br/>r.avg_region_sales AND m.region_profit_margin &gt;<br/>r.avg_region_margin THEN 'Low Sales, High Profitability'</p> <p>ELSE 'Weak Performer' END AS<br/>Evaluation</p> <p>FROM</p> <p>metrics AS m</p> <p>CROSS JOIN</p> <p>region_averages AS r</p> <p>ORDER BY</p> <p>m.total_region_sales DESC</p> |                                                                                                                                                                                                                           |
| 8 | Which sales method generates the highest profit margins and how do their sales contributions compare? | <p>SELECT</p> <p>Sales_Method,</p> <p>ROUND(100 *<br/>SUM(Operating_Profit)/SUM(Total_Sales), 2) AS<br/>Profit_Margin,</p> <p>SUM(total_sales) AS "Total_Sales",<br/>ROUND((100.0 *<br/>SUM(Total_Sales))/(SELECT SUM(Total_Sales)<br/>FROM Sales), 2) AS Contribution_Percent</p> <p>FROM</p> <p>Sales</p> <p>GROUP BY</p> <p>Sales_Method</p> <p>ORDER BY</p> <p>Profit_Margin DESC</p>                                                                                                             | <ul style="list-style-type: none"> <li>Sales conducted via the online platform generates the highest profit margin at 43.48% while also contributing the most to total sales numbers at 37.42%</li> </ul>                 |
| 9 | Which months generated the highest sales and operating profits, and are there any seasonal trends?    | <p>SELECT</p> <p>strftime('%m', Invoice_Date) AS Month,</p> <p>SUM(Total_Sales) AS "Total Sales",<br/>SUM(Operating_Profit) AS "Operating Profits"</p> <p>FROM</p> <p>Sales</p> <p>GROUP BY</p> <p>Month</p>                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Revenue and profits are at their highest during the months of July, August, and December. Revenue and profits are at their lowest during the months of February, March,</li> </ul> |

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|  |  |  | <p>October, and November.</p> <ul style="list-style-type: none"><li>• This shows that revenue and profits are at their peaks during the summer months and around the year end holidays.</li></ul> |
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